



Furnace Gas Pressure Check

Field Standard Operating Procedure | Brackett Heating, Air & Plumbing

BEFORE YOU START

The unit rating plate and the manufacturer's installation instructions are the authority on every number in this document. The values here are typical industry figures for orientation only. When they disagree with the rating plate, the rating plate wins.

Do not leave a safety control bypassed. Do not operate equipment you have reason to believe is unsafe. If you smell gas: do not operate switches, shut off the gas, evacuate, and call the gas utility from outside.

1. When to check

- New installation startup and every annual maintenance.
- Any no-heat, short-cycle, delayed-ignition, flame-rollout, or lockout call.
- After a fuel conversion, an orifice change, or a high-altitude kit install.
- Any complaint of insufficient heat, especially when other gas appliances are running.
- After the gas utility works on the meter, service, or regulator.

2. Safety

- Safety glasses. Personal CO monitor on and in the work area.
- No smoking, no open flame, no ignition sources. Ventilate the space.
- Close the manual gas shutoff before removing any pressure tap plug or backing out a tap set screw.
- The furnace must run for these readings. Line voltage, the inducer, and the blower are live. Keep hands and leads clear of moving parts.
- Leak check every tap you open with an approved leak detection solution before you leave. Never test for leaks with a flame.
- Gas piping, venting, and appliance work must comply with NFPA 54 / the National Fuel Gas Code and all local codes.

3. Tools

- Digital manometer, 0 to 35 in. W.C. range (a U-tube manometer is acceptable).
- 3/16 in. ID manometer hose plus barbed 1/8 in. NPT adapters.
- 3/32 in. hex key or a small flat screwdriver for gas valve tap set screws.
- Approved leak detection solution.
- Combustion analyzer and two thermometers (temperature rise verification).
- Jumper leads for staging a two-stage board.

4. Typical target pressures

Confirm against the rating plate before adjusting anything. Manifold tolerance is commonly plus or minus 0.3 in. W.C.

Measurement	Natural Gas	LP / Propane	Notes
Inlet, static (burners off)	Within the plate's max, commonly 7 to 10.5 in. W.C.	Within the plate's max, commonly 11 to 13 in. W.C.	Must never exceed the valve's stated maximum inlet pressure.
Inlet, running (full fire)	Commonly 5 to 10.5 in. W.C.; 7 in. W.C. nominal	Commonly 11 to 13 in. W.C.; 11 in. W.C. nominal	Most manufacturers set a hard minimum near 4.5 in. W.C. (NG) and 11 in. W.C. (LP).
Manifold, single stage	3.5 in. W.C.	10.0 in. W.C.	The number printed on the rating plate is the target.
Manifold, low fire (two stage)	Commonly 1.6 to 1.8 in. W.C.	Commonly 4.5 to 6.0 in. W.C.	Varies widely by model. Plate governs.
Manifold, high fire (two stage)	3.5 in. W.C.	10.0 in. W.C.	Set high fire first, then low fire.
Static-to-running drop	1 in. W.C. or less	1 in. W.C. or less	A larger drop points to a supply restriction, not the furnace.

Altitude: above 2,000 ft many manufacturers require a derate, a different orifice, or a high-altitude kit. Check the installation instructions before judging a reading good or bad.

5. Inlet (supply) pressure

This proves the gas supply can feed the appliance under load. Check it before you touch the manifold regulator.

- 1 Turn the furnace off at the thermostat and the service switch.
- 2 Close the manual gas shutoff upstream of the gas valve.
- 3 Find the inlet tap on the gas valve, marked IN or INLET. Back the set screw out one full turn (do not remove it), or remove the 1/8 in. NPT plug.
- 4 Connect the manometer hose to the tap. Zero the manometer.
- 5 Open the manual gas shutoff. Record the **static** pressure with the burners off.
- 6 Restore power, call for heat, and let the furnace reach full fire.
- 7 Record the **running** pressure at full fire.
- 8 Fire every other gas appliance in the building at the same time: water heater, range, dryer, second furnace, pool heater. Record this **worst-case** running pressure.
- 9 Compare all three readings to the rating plate. The drop from static to worst-case running should be 1 in. W.C. or less.

IF INLET PRESSURE IS OUT OF SPEC, STOP HERE

Do not adjust the manifold regulator to compensate for a supply problem. Correct the supply first, then re-check the manifold. Adjusting the valve to mask low supply produces a furnace that fails the moment another appliance fires.

6. Manifold pressure, single stage

- 1 Turn the furnace off and close the manual gas shutoff.
- 2 Find the outlet tap on the gas valve, marked OUT or MANIFOLD, or the tapped port on the manifold pipe. Back the set screw out one turn, or remove the plug.
- 3 Connect the manometer and zero it.
- 4 Open the gas, restore power, and call for heat.
- 5 Let the furnace run 3 to 5 minutes so the reading stabilizes. Read at full fire.
- 6 Compare to the manifold pressure printed on the rating plate.

7. Two-stage and modulating

- Two-stage valves have two adjustment screws under separate caps, usually marked HI and LO. **Set high fire first, then low fire** - on many valves the high-fire setting shifts the low-fire reading.
- To force stages manually, isolate the thermostat and jumper R to W1 for low fire, then R to W1 and W2 for high fire. Many boards also have a setup or component-test mode that will do this for you; use it where available.
- Modulating furnaces: use the manufacturer's field-test or manual firing-rate mode and step through the rates. Do not attempt to infer the setting from a single reading.
- Record both low-fire and high-fire readings on the invoice.

8. Adjusting the regulator

- 1 Remove the regulator cap screw on the gas valve.
- 2 Turn clockwise to **increase** pressure, counterclockwise to **decrease**.
- 3 Move in small increments, roughly a quarter turn, and let the reading settle between adjustments.
- 4 Reinstall the cap screw. The regulator will not control correctly with the cap off.
- 5 Never set manifold pressure above the rating plate value to get more heat out of the furnace. Over-firing cracks heat exchangers and makes CO.
- 6 If you cannot reach the target, the problem is upstream supply or the valve itself. Do not keep turning.

9. Shutdown and verification

- 1 Turn the furnace off and close the manual gas shutoff.
- 2 Remove the hose. Reinstall the plug or snug the set screw. These are brass - snug, not gorilla-tight.
- 3 Open the gas and restart the furnace.
- 4 **Leak check every tap you opened** with approved solution. This step is not optional.
- 5 Verify temperature rise falls within the range on the rating plate.
- 6 Run a combustion analysis and record CO.
- 7 Reinstall all doors and panels. The furnace will not vent correctly without them.

10. Reading the results

Symptom	Likely cause	Next step
Static inlet OK, drops hard under load	Undersized or long gas pipe run; undersized or failing primary regulator; partially closed valve; plugged line filter; LP tank low or poor vaporization in cold weather	Size the pipe against the length and total connected load. Check the LP tank level and ambient temperature.
Static and running both low	Utility supply pressure, meter sizing, or the service regulator	Verify at the meter outlet. Call the gas utility if the supply itself is short.
Inlet good, manifold low	Gas valve regulator out of adjustment or the valve is failing	Adjust per section 8. If it will not come up, replace the valve.
Manifold high	Regulator misadjusted, incomplete fuel conversion, or a failed valve	Confirm the conversion kit matches the fuel. Verify the orifices were changed.
Pressures correct, still underperforming	Wrong orifice size, missed altitude derate, airflow restriction, or a dirty heat exchanger	Check temperature rise and static pressure on the air side.

11. Record on every call

Inlet static · inlet running · inlet with all appliances firing · manifold low fire · manifold high fire · temperature rise · CO (ppm, air free) · fuel type · model and serial · date and technician.